Questionnaire in research on Applying Software Engineering for Improving the Development of Data Science Application (Prolific)

MU-CIRB	**Mahidol University Central Institutional Review Board (MU-CIRB)**	**Version Date 10/07/2020**
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Participant Information Sheet for answering the Questionnaire

☐ Original ☐ Revise No. .... Revise Date ..../.........../...........

Dear Respondents,

My name is Dr. Chaiyong Ragkhitwetsagul, the principal investigator, would like to invite you to participate in my research entitle **"Applying Software Engineering for Improving the Development of Data Science Application".** This research project aims to study the challenges that data scientists are facing and whether an application of software engineering methodologies and best practices can help alleviate such challenges.

You are invited to participate in this research project because you are working as a data science practitioner. There will be approximately 140 participants, and the research project will last for 12 months.

If you decide to participate in the research project, you will go through these procedures:

You are invited to answer the self-administered questionnaire. The questionnaire consists of 20 questions and it will take **about 15-30 minutes to complete this questionnaire.** On completion, please return the questionnaire by submitting this online form.

In filling out questionnaires, the likely risks include uneasiness or discomfort due to some questions and in filling out questionnaires; the likely risks include stress due to some questions. In those cases, you have the right not to reply.

As a participant of this study, there will be financial compensation of 50 THB (gift voucher) given to you after you submit the survey. If relevant information arises about benefits and risks of the research project, I will inform you immediately and without concealment.

If you have any questions about the research procedures, you can contact Dr. Chaiyong Ragkhitwetsagul, Faculty of ICT, Mahidol University, Telephone number: 089-1763372

Your private information will be kept confidential, it will not be subject to an individual disclosure, but will be disseminated as part of the overall results. Individual information may be examined by groups of persons e.g. funding organizations, ethics committee, etc.

You have the right to withdraw from the project at any time without prior notice. And the refusal to participate or the withdrawal from the research project will not at all effect on the treatment that you will receive.

On the condition that you are not treated as indicated in this information sheet, you can contact the Chair of Mahidol University Central Institutional Review Board (MU-CIRB) at the office of MU-CIRB, Research Administration Division, Office of the President, Mahidol University, Tel 66-2-8496224-5 and Fax 66-2-8496224.

Thank you very much for your participation

APPROVED
Mahidol University
Central Institutional
Review Board (MU-CIRB)
Protocol No. 2022/173.2206

Digitally signed by Mahidol University Central Institutional Review Board
DN: c=TH, o=Mahidol University, ou=01, cn=Mahidol University Central Institutional Review Board

* 1. I confirm that I am above 18 years old and I give my consent to participate in this study.

◯ Yes

◯ No

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* 1. Please select your country

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Part 1 : Demographics

* 1. Please indicate your work experience in data science

*"Work" means working in an organization or freelance work with salary or renumeration. It does not include self-learning

◯ No experience

◯ Less than 1 year

◯ 1-5 years

◯ 6-10 years

◯ More than 10 years

* 2. Please indicate your work experience in software engineering

*"Work" means working in an organization or freelance work with salary or renumeration. It does not include self-learning

◯ No experience

○ Less than 1 year

○ 1-5 years

○ 6-10 years

○ More than 10 years

* 3. Your education level

○ Lower than undergraduate

○ Undergraduate

○ Master's

○ PhD

* 4. Your current position

○ Data scientist

○ Data analyst

○ Data engineer

○ Machine learning engineer

○ Project manager

○ Programmer / Software developer

○ Other (please specify)

* 5. Type of your organization

- ◯ Software house
- ◯ In-house data science/data team
- ◯ Data consultant
- ◯ Other (please specify)

* 6. Size of your organization

- ◯ 0 - 9
- ◯ 10 - 99
- ◯ 100 - 999
- ◯ 1,000 - 9,999
- ◯ 10,000 - 99,999
- ◯ 100,000 or above

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Part 2: Awareness and Adoption of Data Science Frameworks or Processes

* 1. Are you aware of / use any data science frameworks or processes?

○ Aware but never use

○ Aware and currently using

○ Not aware

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* 1. Which data science framework or data science process that you are using?

- CRISP-DM
- Microsoft TDSP
- Other (please specify)

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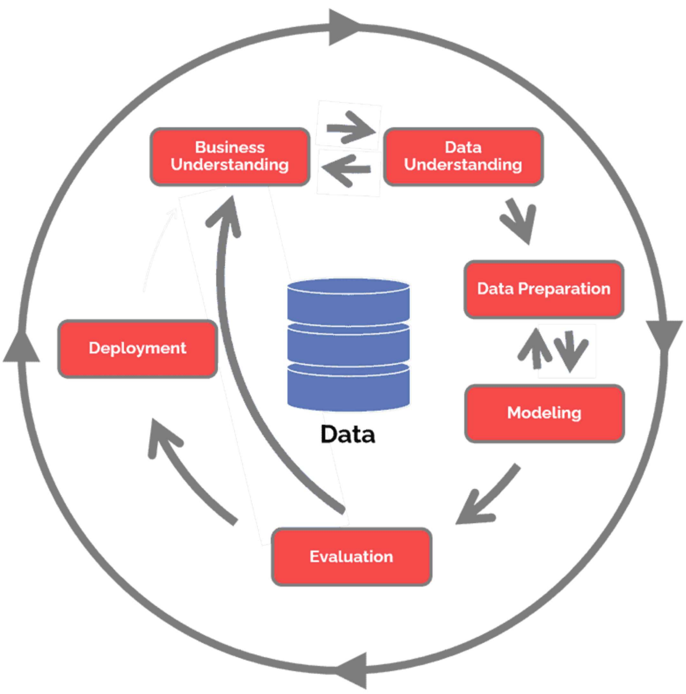
Part 3: Challenges in Data Science During the Project Development

Please spend some time to understand the diagram below showing 6 phases in a data science project development based on the CRISP-DM process model.

The 6 phases have the following definitions.

1. Business Understanding (BU) = Understanding the business objectives or the problems of the client
2. Data Understanding (DU) = Understanding the data used for the project
3. Data Preparation (DP) = Preparing the data to be appropriate for the project (e..g, data cleansing)
4. Modeling (MOD) = Creating a model(s) from the data
5. Evaluation (EVA) = Evaluating whether the model actually answers the client's problem or objectives.
6. Deployment (DPM) = Deploying the model for actual use



1. Please ranking how frequently in each phase in data science projects that you often follow (0 = Never / 5 = Very Frequently)

	0 - Never	1 - Very Rarely	2 - Rarely	3 - Occasionally	4 - Frequently	5 - Very Frequently
Business Understanding	○	○	○	○	○	○
Data Understanding	○	○	○	○	○	○
Data Preparation	○	○	○	○	○	○
Modeling	○	○	○	○	○	○
Evaluation	○	○	○	○	○	○
Deployment	○	○	○	○	○	○

2. How frequently are you face the challenges you during the development of data science projects? (0 = Never / 5 = Very Frequently)

	0 - Never	1 - Very Rarely	2 - Rarely	3 - Occasionally	4 - Frequently	5 - Very Frequently
High learning curve of business [BU]	○	○	○	○	○	○
Misunderstanding between data scienc team and the users [BU]	○	○	○	○	○	○
New and unplanned features requested [BU]	○	○	○	○	○	○
Data cleansing [DP]	○	○	○	○	○	○
Data inconsistency or data behavior shift, data drift [DP]	○	○	○	○	○	○

	0 - Never	1 - Very Rarely	2 - Rarely	3 - Occasionally	4 - Frequently	5 - Very Frequently
Incomplete data [DP, DU]	◯	◯	◯	◯	◯	◯
High learning curve and exploration on the data [DU]	◯	◯	◯	◯	◯	◯
Data assertion problem and data observability and monitoring [DPM]	◯	◯	◯	◯	◯	◯
Model monitoring [DPM]	◯	◯	◯	◯	◯	◯
Fixing bugs in model [MOD]	◯	◯	◯	◯	◯	◯

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Part 4 : Challenges in Software Engineering During the Project Development

1. What are the challenges related to software engineering that you faced during the development of data science projects?

- [ ] No documentation for maintenance
- [ ] Many stakeholders in the project cause difficulties in managing the project
- [ ] No integration and load testings
- [ ] Requirement changes
- [ ] Requirement gathering problems such as incomplete or incorrect requirements
- [ ] No version control system
- [ ] Maintenance and reuse of source code
- [ ] Other (please specify)

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Part 5 : Tools

1. What are the tools that you use for developing data science projects? Please indicate the status of each tool (i.e., used to use, currently using, plan to use in the future) and also select the data science phase that the tool is being used.

	Used to use	Currently using	Plan to use in the future
Apache Airflow™	○	○	○
Databricks	○	○	○
Dataflow	○	○	○
Jupyter Notebook	○	○	○
Streamlit	○	○	○
TensorFlow Serving	○	○	○
Torchserve	○	○	○
Coltjob	○	○	○
Python packages and libraries (ex.	○	○	○

	Used to use	Currently using	Plan to use in the future
Pandas, scikit-learn, numpy)			
Big Query	○	○	○
Blacklink	○	○	○
Cloudera Data Science Workbench (CDSW)	○	○	○
Data Built Tool (DBT)	○	○	○
Google Colab	○	○	○
Apache Hive	○	○	○
i-Source	○	○	○
Impala	○	○	○
Jupyter Lab	○	○	○
MLFlow	○	○	○
RapidMiner	○	○	○
Amazon SageMaker	○	○	○
Tableau	○	○	○
R libraries and packages	○	○	○
SAS	○	○	○
Scala	○	○	○
SQL	○	○	○
Tkinter	○	○	○

	Used to use	Currently using	Plan to use in the future
PyCharm	◯	◯	◯

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1. Please give a score of the software engineering tools, which are categorized as shown below, that you are currently using in your data science projects (5 = Extremely useful, 1=Extremely not useful)

	1	2	3	4	5	Not applicable/I don't use it
Software Requirement Tools such as Requirements Modeling Tools (Using for eliciting, analyzing, specifying, and validating requirements) and Requirement Traceability Tools for example Siemens Polarion, DOORS DNG (IBM)	○	○	○	○	○	○
Software Design Tools such as tools for create	○	○	○	○	○	○

	1	2	3	4	5	Not applicable/I don't use it
and analyze software designs for example Visual Paradigm, IBM Rational Software Architect Designer						
Software Construction Tools such as Program Editiors, Compilers and Code Generators, Interpreters and Debuggers for example VS Code, PyCharm, IntelliJ	○	○	○	○	○	○
Software Testing Tools such as Test Generators, Test Execution Frameworks, Test Evaluation Tools, Test Management Tools and Performance Analysis Tools for example Selenium, Postman, Unit testing frameworks.	○	○	○	○	○	○
Software Maintenance Tools such as	○	○	○	○	○	○

	1	2	3	4	5	Not applicable/I don't use it
Comprehension Tools (for example Visualization tools such as animators and program slicers) and Reengineering Tools for example SonarQube						
Software Configuration Management Tools such as Defect, Enhancement, Issue, and Problem-Tracking Tools, Version Management Tools and Release and Build Tools for example GitHub, Gitlab.	○	○	○	○	○	○
Software Engineering Management Tools such as Project Planning and Tracking, Risk Management and Measurement Tools for example Jira, Trello.	○	○	○	○	○	○

	1	2	3	4	5	Not applicable/I don't use it
Software Engineering Process Tools such as Process Modeling Tools, Process Management Tools, Integrated CASE environments and Process-centered Software Engineering Environments for example Signavio.						
Software Quality Tools: Review and Audit and Static Analysis Tools. It also includes Inspection Tools such as SonarQube.						
Knowledge Management Tools: tools that support the knowledge lifecycle processes, such as the creation or distribution of knowledge (for instance a WIKI tool) such as Google Docs, Microsoft SharePoint.						

	1	2	3	4	5	Not applicable/I don't use it
Virtual Meeting Tools: tools which principally permit communication among distributed teams. Examples of these are videoconference tools, virtual room tools, etc such as Zoom, WebEx, Microsoft Teams.	○	○	○	○	○	○
Socio-Cultural Tools such as tools that support the interaction like social networks and social network analysis tool for example LINE, Facebook, Discord and Twitter.	○	○	○	○	○	○
Miscellaneous Tools such as Meta-tools or Integration Tools that helping to manage the complex system for example Jenkins, GitHub Actions	○	○	○	○	○	○

Questionnaire in research on Applying Software Engineering for Improving the Development of Data Science Application (Prolific)

1. What are the methodologies or best practices in software engineering that you use for developing data science projects? Please indicate the status of each tool (i.e., used to use, currently using, plan to use in the future) and also select the data science phase that the tool is being used.

	Used to use	Currently using	Plan to use in the future
Scrum or Kanban board	◯	◯	◯
Stand-up meeting	◯	◯	◯
Releases in sprints	◯	◯	◯
Code review	◯	◯	◯
Coding convention	◯	◯	◯
DevOps pipelines (e.g. CI/CD)	◯	◯	◯
Effort estimation (e.g. storypoint estimation)	◯	◯	◯
Requirement analysis (e.g.	◯	◯	◯

	Used to use	Currently using	Plan to use in the future
functional vs. non-functional requirements)			
Testing - define exit criteria	◯	◯	◯
Integration testing	◯	◯	◯
Model testing	◯	◯	◯
User acceptance testing (UAT)	◯	◯	◯
Unit testing	◯	◯	◯
Use case analysis	◯	◯	◯
Containerization (e.g. Docker)	◯	◯	◯

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1. Do you have any software engineering methodologies or best practices that you want or plan to use in the future but are not listed above?

2. What are the benefits from adopting software engineering methodologies and best practices in data science projects?

- [ ] Getting team's suggestions during stand-up meetings
- [ ] Help following the plan
- [ ] Increase efficiency in every step
- [ ] Increase test coverage
- [ ] Reduce bugs before giving the project to the users
- [ ] Tidy code and high readability
- [ ] Easy transfer of the project

☐ Other (please specify)

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* 1. What is your dream tool for data science? This will be the tool that does not currently exist and you want to have it so that you can fix your current challenges or issue. Please give a short explanation of the tool

* 2. What is your Prolific ID?

* 3. Copy the code C14PY3O4 and submit it on Prolific to indicate that you have completed the study.

☐ Yes, I have copied the code and submitted in in Prolific.

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